A rare case of pachydermoperiostosis associated with blepharoptosis and floppy eyelids

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Pachydermoperiostosis (PDP) is a multisystem disorder of mesenchymal origin. It is a form of hypertrophic osteoarthropathy. The typical clinical features include pachydermia, cutis verticus gyrata, digital clubbing, and periostosis. Patients present with mechanical ptosis resulting from markedly thickened eye lids. Floppy eye lids have rarely been reported in association with PDP. We describe a rare case of PDP associated with ptosis and floppy eye lids in an adult male. Meibomian gland dysfunction was documented by meibography. The patient underwent bilateral upper lid wedge resection, lateral tarsal strip, tarsectomy and external levator advancement with good cosmetic outcome following surgery.

Key words: Floppy eyelid, hypertrophic osteoarthropathy, meibography, pachydermoperiostitis

Pachydermoperiostosis (PDP) or Touraine-Solante-Gole syndrome is a rare idiopathic mesenchymal disorder characterized by the presence of pachydermia (thickening of skin or elephant skin), digital clubbing and periostosis usually seen in adolescent males.[1,2] Floppy eyelids has rarely been described in association with PDP.[3] We report the clinical features and surgical outcome of a case of PDP with ptosis, floppy eyelids, and meibomian gland dysfunction documented by meibography.

Case Report

A 32-year-old male from Nepal presented to the oculoplasty clinic with complaints of redness and swelling of both the upper eyelids followed by progressive drooping for the last 5 years. There was no history of diurnal variation, double vision, trauma or any previous surgery. The patient also gave history of recurrent episodes of painful, swollen joints (hip, knee and ankle). Similar ocular as well as orthopaedic complaints were seen in one of his paternal uncles and grandfather.

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On examination visual acuity was 20/20, N6 in both eyes. Marginal reflex distance-1 was −1 mm in both eyes with excellent levator action. Both upper lids were thickened and markedly floppy [Fig. 1b and c]. The thickness was more pronounced on the lateral aspect and showed eversion. Patient was orthophoric with good bells and normal extra ocular movements. There was pachydermia of forehead and scalp, with deep forehead furrows [Fig. 1b]. The fingers and toes of the patient were broad with mild clubbing [Fig. 1d]. Meibomian gland orifices were blocked and appeared flat. Meibography revealed severe dropout of meibomian glands with marked distortion in the architecture [Fig. 2]. None of these abnormalities were seen in his old childhood photo [Fig. 1a]. Patient was diagnosed to have PDP.

Bone scan revealed osteopenia of the hip joint. Patient was advised to start oral residronate and osteocalcium by orthopaedist, and his joints showed marked symptomatic improvement. The patient also underwent bilateral upper lid wedge excision with lateral tarsal strip with external levator advancement. Per operatively the tarsal plates were found to be hypertrophied with increased width and height [Fig. 1e]. The extra tarsus (around 5 mm) was excised. A good correction of ptosis and lid laxity was noted in the immediate postoperative period [Fig. 3]. Histopathological examination of the resected tarsal tissue showed fibrosis, sebaceous gland hyperplasia, mucin deposition and nonspecific chronic inflammation, which are typically seen in PDP [Fig. 4].

Patient was started on aggressive lubricants in view of meibomian gland dysfunction and was asked to review after 6 weeks. However the patient failed to return for review.

Discussion

PDP was first described by Nikolaus Friedrich in 1868 as a form of acromegaly. Later on in 1935 Touraine, Solente, and Gole identified the condition to be quite distinct and described it as syndrome of pulmonary hypertrophic osteoarthropathy.[4] Hypertrophic osteoarthropathy can be either primary or secondary with similar clinical features including pachydermia, cutis verticus gyrata, digital clubbing, and thickening of long bones (periostosis).[5] The secondary form is associated with an underlying pathology, is almost always painful and rapidly progressive. Primary hypertrophic osteoarthropathy is also known as PDP. It is seven times more common in males, inherited in autosomal dominant fashion with variable expression.[2] The onset of PDP is usually in the adolescence, wherein there is clubbing and broadening of limbs resulting from diffuse periostial ossifications. The changes in PDP progress over the next 5–20 years and then stabilize to remain unchanged throughout life.[6]

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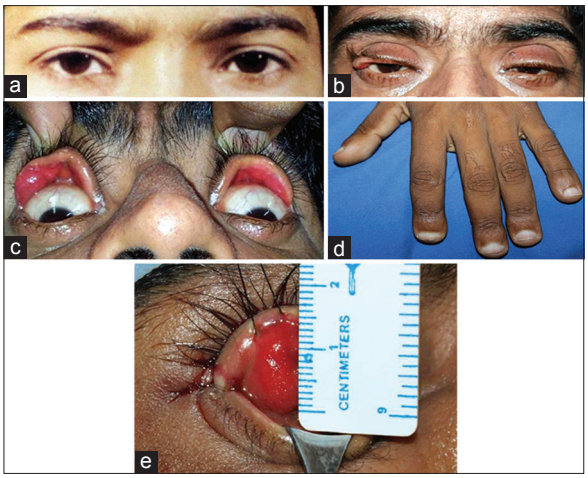


Figure 1: (a) Childhood photograph of the patient showing normal appearing eye lids. (b) External photograph showing pachydermia of forehead, face, and eye lids with mechanical ptosis and deep forehead furrows. (c) External photograph showing floppy eye lids. (d) External photograph showing digital clubbing. (e) Elongated upper tarsal plate



Figure 3: Postoperative (day 1) external photograph showing good correction of ptosis and floppy eye lids

Ptosis, caused by thickening of the eyelids, has been associated with PDP; but floppy eyelids have rarely been described.[3] We also documented the presence of severe distortion in meibomian gland architecture, a feature not previously described, by meibography. Other reported ocular features include corneal leukoma, presenile cataract, and macular dystrophy.[3] PDP needs to be differentiated from similar conditions like acromegaly, leprosy, syphilis, and Paget's disease.[5] There are no hormonal or hematological abnormalities seen in PDP. Development of eyelid thickening in PDP has been attributed to sebaceous gland hyperplasia and mucin deposition.

Medical management includes nonsteroidal anti-inflammatory drugs, corticosteroids, tricyclic antidepressants, resedronate, pamidronate, and tamoxifen citrate to control arthritis.[7] Surgical management of PDP is challenging and mostly combines horizontal lid shortening with ptosis correction, frontal rhytidectomy, and brow lift.[6] These can be done either in a staged fashion or in a single

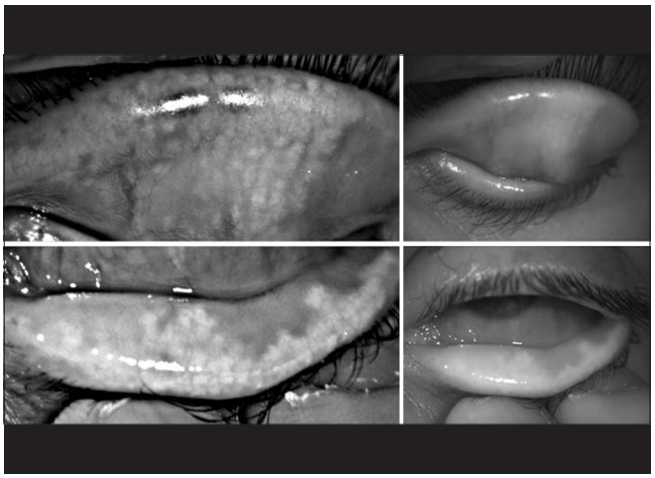


Figure 2: Meibography showing distorted architecture of meibomian glands with severe dropout

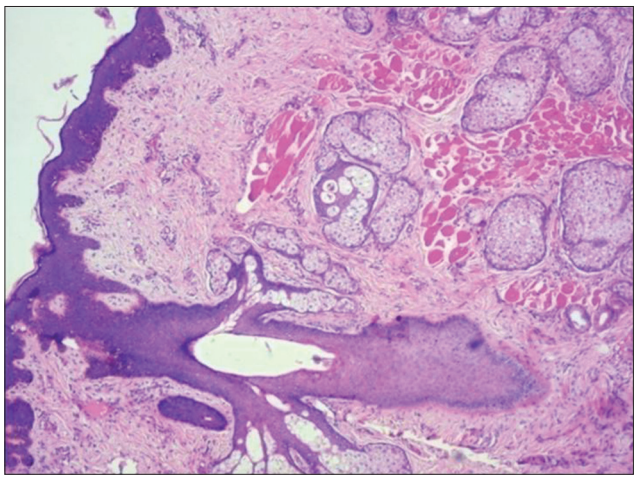


Figure 4: Microphotograph (H and E; ×20) showing sebaceous gland hyperplasia, fibrosis, and nonspecific chronic inflammation

sitting.[4] Berdia *et al*. advocated staged bilateral upper eye lid wedge resection followed by external levator advancement.[4] However, we feel that the lid laxity and the blepharoptosis can be addressed simultaneously and gratifying results can be obtained. As these patients can present initially with ocular symptoms, it is extremely important for an ophthalmologist to recognize this disorder. Various systemic associations, some of them malignant (gastric adenocarcinoma).[8] Early diagnosis and referral to appropriate specialties (dermatology, rheumatology) is crucial for management and prevention of disabilities like deafness, kyphosis, and osteonecrosis of the femoral head.[9]

PDP is a rare idiopathic inherited disorder of mesenchymal origin. It can be associated with mechanical ptosis, floppy eye lids, and meibomian gland dysfunction. Characteristic morphological appearance and body habitus along with typical radiological features help in clinching the diagnosis.

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Conflicts of interest

There are no conflicts of interest.

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